University of Tulsa

Gas and Oil Investment Memo

Samuel Abiel, Ashlynn Comstock, Jariah Eyachabbe, Luna Wolfe
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Dr. Cayman Seagraves
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Executive Summary

Findings overview:

This capstone studies how changes in energy markets, like oil and natural gas shocks, affect consumer inflation (higher prices for goods and services). It focuses on whether countries that rely heavily on imported energy and affect differently than countries that are more energy independent. This is important because energy prices often rise suddenly and can impact the global economy.

Our results showed three main points. First, natural gas import shocks had only a very small effect on inflation overall, meaning energy shocks did not strongly increase inflation in the average country. Second, countries that depend more on energy imports actually seemed to have weaker inflation responses than less dependent countries. This may be because those countries are more prepared through hedging, subsidies, or reduced demand. Third, when comparing countries before and after the 2022 crisis, there was no strong evidence that energy-dependent countries performed significantly worse.

There are some limits to the study. We measured shocks using import volume changes instead of global energy prices, which may miss part of the true effect. We also used yearly inflation data with monthly energy data, which can reduce accuracy. Other factors like interest rates, exchange rates, and government policies were not fully included. Even so, the results suggest that countries handle energy shocks differently, and this deserves more study.

Methodology

For this project, we used a variety of data sets to get results that were relatively unbiased. We utilized the Joint Organization Data Initiative (JODI Oil & Gas Data) from the Open dataset catalog to reflect oil prices in the US and other countries. Jodi Oil and Gas data also reflects global availability to help us evaluate shocks affecting the global market. The second set we used was sourced from the Federal Reserve Economic Data (FRED) which covers CPI inflation, policy rates, and industrial production. We are able to cross reference global shocks and policy rates and consider how industrial production is correlated with CPI inflation. Finally, we used the World Data Bank to add energy import dependance by country.

Our three Datasets combined for a sample of 323359 rows with 19 variables. During Milestone 1 we spent a great deal of time cleaning the data, we dropped any rows with missing values, standardized dates column names to be conventionally the same across all datasets, and we dropped any empty columns. Date was standardized with YYYY-MM-DD format across all datasets. The Consumer Price Index (CPI) is the measurement of inflation. Policy-Rate is the Central Banks interest rate that guides borrowing costs in the economy at a given time. Industrial Production is an index measuring output from mines and factories which stands for real economic activity in our analysis. There are many more variables included in our regression, but these are the necessary definitions needed to understand our figures and the findings of our study.

We created two main models for this study, the first being the two-way fixed effects panel regression. This model utilizes country fixed effect and year fixed effect to minimize bias from any omitted factors we could not include stemming from institutional, geographic, and global inflationary pressure. The other model we used with Difference-in-Differences contrasts high-dependance countries and low-dependance countries with how much they import energy since 2022.

Results

Table 1 reports the main Fixed Effects regression results, Table 2 reports the alternative DiD model, Figure 1 shows the relationship between industrial production and the policy rate over time, and Figure 2 evaluates the Fixed Effects model using a residuals-versus-fitted diagnostic plot.

Table 1:

Term	**Standard Error**	**Model (1) Fixed Effects**
Gas import shock (t-1)	0.0009	-0.0001
Shock × high energy dependence	0.0010	-0.0005
GDP growth	0.0756	-0.2360
Imports (% GDP)	0.0587	0.1198

Country fixed effects	Yes
Year fixed effects	Yes
Clustered standard errors	Country
Observations	430
R-squared	0.1241

Table 1 presents the main Fixed Effects of regression results. The coefficient on **Gas import shock (t-1)** is **-0.0001**, meaning a 1 percentage-point increase in the previous period's gas import shock is associated with a **0.0001 percentage-point decrease in inflation**, controlling for country and year fixed effects. However, this effect is very small and not statistically significant.

The interaction term, **Shock × high energy dependence**, is also small and not statistically significant, suggesting that high-energy-dependence countries do not show a clearly different inflation response to gas import shocks in this model. The strongest findings are for **GDP growth**, which is negative and significant, and **Imports (% GDP)**, which is positive and significant. Overall, the model suggests that gas import shocks alone are weak predictors of inflation, while broader macroeconomic and trade factors are more meaningful.

Table 2:

Term	**Standard Error**	**Model (1) Fixed Effects**
Gas import shock (t-1)	0.0004	-0.0005
High dependence × post-2022	1.0464	-0.2502
High energy dependence	1.1155	3.4994
Post-2022	0.6866	1.7571
GDP growth	0.0785	-0.2469
Imports (% GDP)	0.0509	0.0990

Country fixed effects	Yes
Year fixed effects	Yes
Clustered standard errors	Country
Observations	477
R-squared	0.7564

Table 2 presents the alternative Difference-in-Differences model. The key DiD variable, **High dependence × post-2022**, has a coefficient of **-0.2502**, meaning high-energy-dependence countries had an estimated **0.2502 percentage-point lower post-2022 inflation shift** than lower-dependence countries, controlling for other factors. However, this result is not statistically significant.

The stronger results are for **High energy dependence** and **post-2022**. High-energy-dependent countries generally had higher inflation, and inflation was also higher after 2022 across the sample. GDP growth remains negative and significant, while Imports (% GDP) remain positive and marginally significant. Overall, the DiD model explains more variation than the Fixed Effects model, but the main DiD interaction does not provide strong evidence of a distinct post-2022 treatment effect.

Figure 1

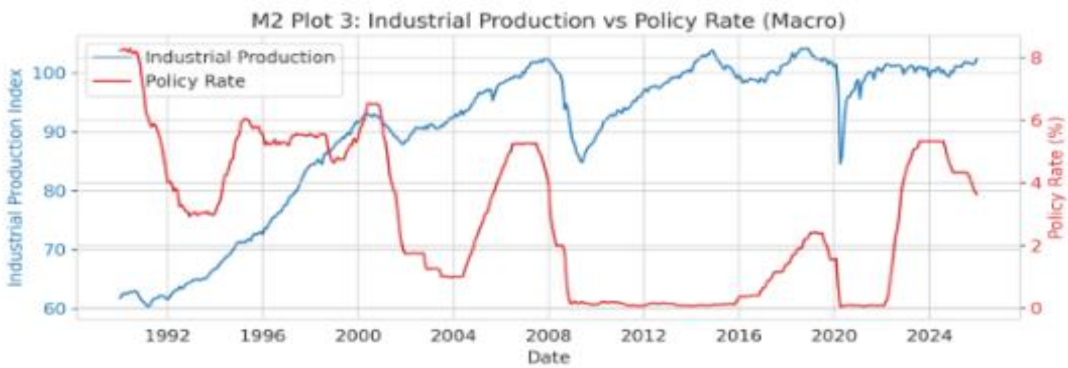

Figure 1 shows industrial production and the policy rate over time. The figure shows that policy rates often fall after major downturns, especially around the 2008 financial crisis and the 2020 COVID shock. This pattern suggests countercyclical monetary policy, where rates are lowered when economic activity weakens.

The figure also supports the broader idea that tighter policy is associated with weaker industrial output. The relationship may appear with a delay, since the lagged correlation evidence shows a stronger negative relationship at longer lags, especially around 12 months. Overall, Figure 1 suggests that higher interest rates are connected with lower industrial production over time.

Figure 2 Diagnostic plot

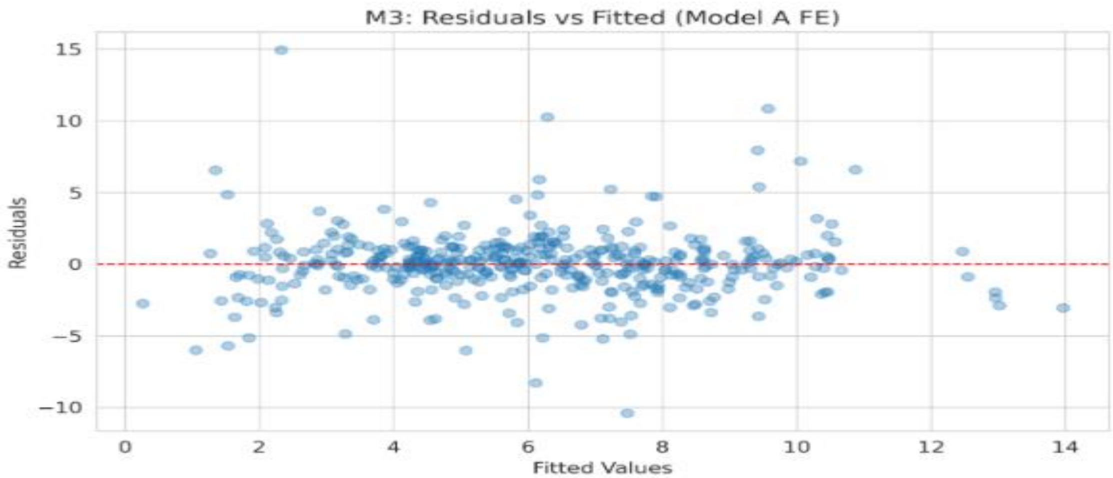

Figure 2 evaluates the Fixed Effects model by comparing fitted values with residuals. The residuals are generally centered around zero, which suggests the model is not consistently overpredicting or underpredicting inflation.

However, the plot also shows outliers and a wider spread at higher fitted values. This suggests possible heteroskedasticity or influential events that the model does not fully capture. These may reflect major shocks such as the 2008 financial crisis, the 2020 COVID shock, or energy-market disruptions. Therefore, the model is useful, but the results should still be interpreted with caution.

Results Summary: Overall, the results show limited evidence that lagged natural gas import shocks directly predict inflation. The main gas shock coefficient and the high-energy-dependence interaction are both small and not statistically significant. However, GDP growth and imports as a percentage of GDP are more meaningful predictors across both models.

Conclusions and Recommendations

During Milestone 1 Our group built a dataset with 323,000 rows and 19 columns that combines data from Federal Reserve Economic Data (FRED), Joint Organizations Data Initiative (JODI), and World Development Indicators (WDI). The dataset is usable but has some sections of missing values and differences across data sources. The issue allowed us to reevaluate the data and handle it with more careful data modeling. In Milestone 2 EDA shows a moderate negative relationship between policy rates and industrial production ($r \approx -0.49$), strongest at about a 12-month lag. Gas flows and demand-import sensitivity vary significantly across countries and flow types. The modeling should account for lagged effects and cross-country differences, while handling outliers, heteroskedasticity, and gaps in macro data. Milestone 3, Two-way FE and DiD specs find small, statistically insignificant gas-import-shock pass-through estimates; interaction terms point directionally to weaker pass-through in high-dependence countries but lack robust significance. Diagnostics (VIF ≈ 2, Breusch-Pagan $p \approx 0.16$) do not reveal major multicollinearity or heteroskedasticity. The evidence suggests modest pass-through and directional heterogeneity, but causal strength is limited by measurement and aggregation choices. These conclusions support our initial question; the panel contains the key variables (import-growth shocks, inflation, country identifiers, dependence indicators), enabling fixed-effect and DiD tests of differential hypothesis. However, our group identified limits being no direct global gas-price series, temporal aggregation, block specific missingness, and potential omitted confounders to attenuate effect sizes and can reduce causal credibility. Our current data allows us to detect directional heterogeneity but are underpowered for precise causal estimates; treat reported pass-through as conservative and exploratory. Short takeaway from an investor perspective; the data only shows modest oil/gas to consumer-inflation and pass-through. This suggests directional heterogeneity across countries, which investors should avoid actions that assume strong, uniform transmission from energy shocks, which Milestone 3 interpretation provides.

References

JODI Oil & Gas Data Open Dataset Catalog (OpenData_rows.csv, id 57; JODI)

FRED Openly available (FRED) via pandas- datareader API

World Bank WDI Open Dataset Catalog (OpenData_rows.csv, id 49; WDI)

Course Materials

GitHub Copilot

Vs Code

AI Audit

Milestone 1: AI helped fetch and organize the source data, convert the binary JODI files into tabular form, clean and standardize datasets, create the consolidated data dictionary, and shape the initial hypotheses. Later M1-related sessions also used AI to identify and remove an all-null column from the final panel and to harden the pipeline and repository cleanup process.

Milestone 2: AI was utilized to make the code creating the heatmap, time series of outcome variable, dual axis plot, and the rest of the milestone 2 visualizations. Additionally, we had AI analyze the readme to create tests so we could ensure we were meeting milestone requirements.

Milestone 3: AI was used to evaluate the project requirements and inform a full modeling script, add two-way fixed effects and DiD specifications, generate diagnostics and robustness checks, produce publication-style tables/figures, and add smoke tests. The audit also notes that the narrative interpretation was checked and corrected when AI initially overstated a directional effect.

Milestone 4: AI drafted the investment memo as an early first-pass document and helped create the JS script that renders the memo to DOCX. The appendix explicitly frames this as a jumping-off point rather than a final deliverable.